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Fa25-bcs-060

Date: September 8,2026

Lab task 01:

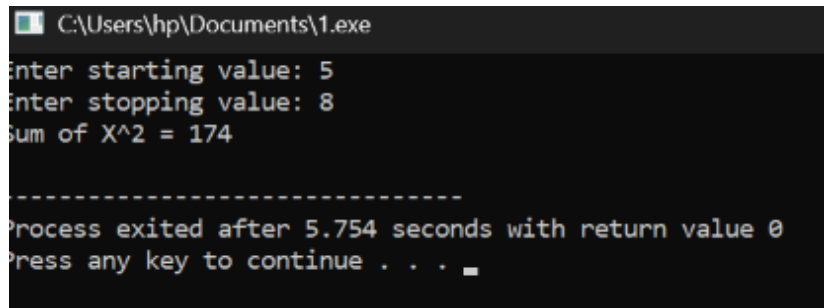
```
#include <iostream>

using namespace std;

int main()
{
    int start, stop;
    int sum = 0;
    cout << "Enter starting value: ";
    cin >> start;
    cout << "Enter stopping value: ";
    cin >> stop;
    for (int x = start; x <= stop; x++)
    {
        sum = sum + (x * x);
    }
    cout << "Sum of X^2 = " << sum << endl;
    return 0;
```

```
}
```

Output:



```
C:\Users\hp\Documents\1.exe
Enter starting value: 5
Enter stopping value: 8
Sum of X^2 = 174

-----
Process exited after 5.754 seconds with return value 0
Press any key to continue . . .
```

Lab task 02:

```
#include <iostream>

using namespace std;

int main() {
    int a[50], n, x, i=0;

    cin >> n;

    for(int j=0; j<n; j++)
        cin >> a[j];

    cout << "Search value: ";

    cin >> x;

    while(i<n) {
        if(a[i] == x) {
            cout << "Value Found";

            return 0;
        }
    }
}
```

```

    }

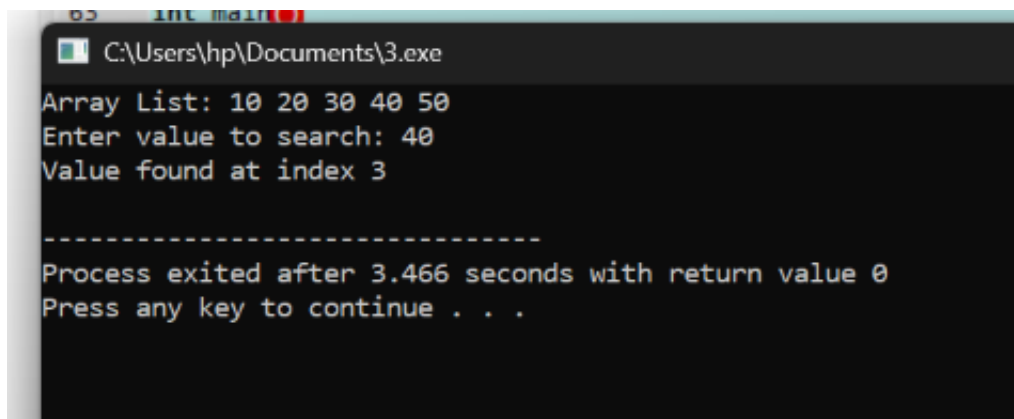
    i++;
}

cout << "Value Not Found";

return 0;
}

```

Output:



```

C:\Users\hp\Documents\3.exe
Array List: 10 20 30 40 50
Enter value to search: 40
Value found at index 3

-----
Process exited after 3.466 seconds with return value 0
Press any key to continue . . .

```

Lab task 03:

```

#include <iostream>

using namespace std;

int main() {

    int a[50], n, x, value, i;

    cout << "Enter size: ";

    cin >> n;

    cout << "Enter values: ";

    for(i = 0; i < n; i++)

```

```
    cin >> a[i];
cout << "Enter value: ";
cin >> value;
a[n++] = value;
cout << "Enter value: ";
cin >> value;
for(i = n; i > 0; i--)
    a[i] = a[i-1];
a[0] = value;
n++;
cout << "Enter value after which to insert: ";
cin >> x;
cout << "Enter new value: ";
cin >> value;
for(i = 0; i < n; i++) {
    if(a[i] == x) {
        for(int j = n; j > i+1; j--)
            a[j] = a[j-1];
        a[i+1] = value;
        n++;
    }
}
break;
```

```

    }
}

cout << "Enter value before which to insert: ";

cin >> x;

cout << "Enter new value: ";

cin >> value;

for(i = 0; i < n; i++) {
    if(a[i] == x) {
        for(int j = n; j > i; j--)
            a[j] = a[j-1];
        a[i] = value;
        n++;
        break;
    }
}

cout << "Array List: ";

for(i = 0; i < n; i++)
    cout << a[i] << " ";

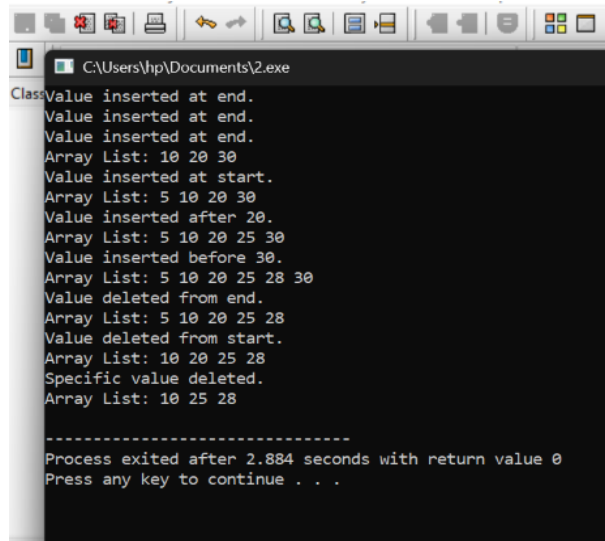
n--;

for(i = 0; i < n-1; i++)
    a[i] = a[i+1];

```

```
n--;  
cout << "\nEnter value to delete: ";  
cin >> x;  
for(i = 0; i < n; i++) {  
    if(a[i] == x) {  
        for(int j = i; j < n-1; j++)  
            a[j] = a[j+1];  
        n--;  
        break;  
    }  
}  
cout << "Final List: ";  
for(i = 0; i < n; i++)  
    cout << a[i] << " ";  
return 0;  
}
```

Output:



```
Class
Value inserted at end.
Value inserted at end.
Value inserted at end.
Array List: 10 20 30
Value inserted at start.
Array List: 5 10 20 30
Value inserted after 20.
Array List: 5 10 20 25 30
Value inserted before 30.
Array List: 5 10 20 25 28 30
Value deleted from end.
Array List: 5 10 20 25 28
Value deleted from start.
Array List: 10 20 25 28
Specific value deleted.
Array List: 10 25 28

-----
Process exited after 2.884 seconds with return value 0
Press any key to continue . . .
```